

Nmap 101: From Zero to Hero in Network Scanning & Discovery

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Welcome, fellow learners! 🙌

Like many of you, I've used Nmap casually for simple scans. But let's be honest: in the world of cybersecurity, "casual" doesn't cut it. I decided to challenge myself to go from Zero to Hero, and I believe the best way to master a skill is to teach it.

But here is the problem: Technical documentation is boring.

Memorizing flags like `-ss` or `-Pn` is painful without context. I hate dry lectures and I bet you do too. That is why we are doing something different in this series.

Instead of a boring tutorial, **we are going to play a game!**

We will turn Nmap into a character: a gritty Private Detective investigating a crime scene. By following his investigation, you won't just memorize commands; you will understand why and how we use them in real world scenarios.

Are you ready to solve the case? Grab your badge. Let's hit the streets.



Mr.Nmap

The Scenario

- Mr. Nmap (He asks the questions).
 - You (You interpret the clues).
 - The Local Network (10.0.2.0/24).
 - A fugitive hiding in an abandoned building (Metasploitable 2)
- -----

STORY TIME, Prologue-CH1:

It was a quiet night in Kali Linux City. The coffee on the desk was cold and the cursor on the terminal was blinking, waiting for orders.

Then, the phone buzzed. It was the Chief (Root(This is you!))

“Detective, we have a situation,” the Chief said. “We have intel on a fugitive hiding somewhere in the 10.0.2.0/24 district. He is an old-school criminal known as ‘The Smiling Bandit’.”

“Do we have an address?” the Detective asked, grabbing his trench coat.

“Negative,” the Chief replied. “We don’t know the exact IP. But we have something better. We know his signature. He is hiding behind a specific service called vsftpd. We just know he is somewhere in that subnet. I need you to go down there, sweep the neighborhood, find out who is home and knock on every door until you find him. Find that service and you find the bandit.

“Consider it done.”

He hung up. He didn’t know the exact IP address yet, but he knew the neighborhood. It was time to hunt.

So in this series, I’ll be breaking down Nmap step by step using my home lab. But first things first:

What is a Port?

Before we start knocking on doors with Nmap, we need to understand the building itself. To explain Ports, I want to use the exact analogy that helped me understand them when I was learning.

Let’s imagine the target machine isn’t a computer. Let’s imagine it as a giant Apartment Complex.

- This is the address of the Apartment Building.
- This is the Apartment Number inside that building.

- This is who lives in the apartment. It's not enough to simply find an open door; as detectives we need to know who is standing behind it.
- This is how we talk to the tenant.

If you send a letter to just the 'Building Address' (IP), the mailman won't know who to give it to. Does it go to Mr. Web Server? Or Mrs. Database?

You need to specify the Apartment Number.

- Usually, the Web Server (HTTP) lives here. It accepts requests for web pages.
- The SSH (Secure Shell) lives here. This is like the maintenance room for admins.
- This is the secure version of Apartment 80 (HTTPS).

So, where does Nmap fit in?

In this analogy, Nmap is the curious Detective scouting the lobby of this building. He doesn't have a master key to just let himself in. Instead, he has to do the legwork: he walks up to every single apartment door and knocks.

- Nmap knocks on Apartment 80. Someone opens the door and says,
- Port 80 is OPEN (Service: Apache).
- Nmap knocks on Apartment 21 (FTP). No one answers, or the door is bricked up.
- Port 21 is CLOSED.
- Nmap knocks on Apartment 22, but a security guard (Firewall) stops it and says
- Port 22 is FILTERED.

This is essentially what we are doing when we run a scan. We are knocking on digital doors to see who is home and what language they speak.

1-THE LAB SETUP

A thief needs a target and a detective needs a crime scene. But we can't just go around scanning google.com or your university's website.

We need a safe, legal environment. You have two options to follow this guide:

If you don't want to install anything, just go to **tryhackme.com**, create a free account and search for the 'Nmap' room. Start the AttackBox and the target machine. You are ready! You don't need to follow their tasks right now; just follow my article using their machine. But you will not be able to do exactly my steps because I have no idea which ports are open or not. But you'll understand it no matter what.

This is what I'm doing. I have set up a virtual lab on my computer using **VirtualBox**. I suggest you to go with this route since you can exactly follow what I'll do.

1. Kali Linux
2. Metasploitable 2 (A machine intentionally built with security holes).

Setup Tip: If you choose this path, make sure to set the Network settings of BOTH machines to NAT Network. Think of ‘NAT Network’ as locking these two machines in a soundproof panic room inside your computer. They can fight each other, but they can’t mess with your home Wi-Fi or be seen by others. If you are stuck on installation, you can just ask ChatGPT or Gemini: “How to set up Kali and Metasploitable 2 in VirtualBox using NAT Network?” or simply copy paste this section. It takes 10 minutes!

2-So It Begins

Every investigation starts with a location. Our Detective (Nmap) is sitting in his car, engine running, ready to work. But he has a problem: He doesn’t know where to drive.

He needs an address to put into his GPS. In the networking world, this “Building Address” is the Target IP Address.

- Your target is easy to find. It is the IP address shown in the green box next to “Active Machine” (e.g., 10.10.x.x).
- We need to find the address of our victim (Metasploitable 2).

First, easy way.

1. Go to your Metasploitable machine window.
2. Type `ifconfig` and hit enter.
3. Look for the `inet addr` (It should look like 10.0.2.x).

That is our Crime Scene. That is the building we are going to investigate. Note that IP down, because we are about to knock on its doors.

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:74:f4:39
          inet addr:10.0.2.3  Bcast:10.0.2.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe74:f439/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:36 errors:0 dropped:0 overruns:0 frame:0
          TX packets:65 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4632 (4.5 KB)  TX bytes:6836 (6.6 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:92 errors:0 dropped:0 overruns:0 frame:0
          TX packets:92 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:19393 (18.9 KB)  TX bytes:19393 (18.9 KB)

msfadmin@metasploitable:~$
```

1.1 ifconfig in metasploitable

`ifconfiginet addr:10.0.2.3`

Phase 1: Is Anyone Home? (Host Discovery)

In my case, the target IP address is 10.0.2.3. Our Detective has the address of the apartment complex, but he needs to be sure: Does anyone actually live there? There is no point in raiding an abandoned building.

So, how do we check? In Nmap, we have a bag full of tricks (Ping scans) to see if the lights are on.

Here are the 3 main ways to check for like:

1. The Polite Ways (ICMP)

These are standard methods. However, strict security guards (Firewalls) often block these.

- **-PE** Imagine walking up to the door and ringing the bell. If they answer, they are home. Simple.
- **-PP** Instead of ringing the bell, you ask the security guard: If he looks at his watch and tells you the time, you know he is real and awake.
- **-PM** Imagine asking the guard: It's a weird question. But if he answers ("From that tree to this wall"), you verify that he exists.

2. The Sneaky Ways (TCP/UDP)

If the polite ways are blocked, we get creative.

- **-PS** We throw a tiny stone at the kitchen window (Port 80) to see if anyone reacts. If they look out, we know they are home.
- **-PA** We yell at someone on the balcony: This is nonsense because we don't know them. Confused, they will yell back: (This is called a RST packet). Their confusion proves they are home.
- **-PU** This works differently. Imagine banging on a garage door that we know is locked and empty. If the owner storms out and yells: (ICMP Port Unreachable), we caught them! Their anger proves they are home.

3. The Neighborhood (ARP)

-PR We don't need to call them or throw stones. Since we are neighbors on the same floor, we just stand in the hallway and yell: "WHO LIVES IN APARTMENT 3?" (Who has this IP?). Since we are close, the target hears us directly and answers: "I do! Here is my ID." (MAC Address). This is the fastest and most reliable method. Nmap does this automatically if it sees you are on the same network.

NOTE: *If you are not root, remember to type **sudo** before these commands.*

We actually cheated a little bit earlier. We looked at the target machine's screen and saw its IP address using **ifconfig**. But in a real investigation, the Detective doesn't get to walk into the criminal's house and ask for his address.

So, let's pretend we don't know the IP yet.

Step 1: Since we are on the same network, we first need to know *our* own address to understand the neighborhood boundaries. Let's check our IP on our attacking machine (Kali): `ifconfig`

Step 2: Now, before we start knocking on doors, we need to know which buildings are actually occupied. We don't want to waste time on empty lots. We will run a Ping Sweep. This tells the Detective to walk down the street and just check who is "home," without knocking on any specific doors yet. `nmap -sn 10.0.2.0/24`

- `/24`: This defines our Target. It tells Nmap to check all 256 addresses from `10.0.2.0` to `10.0.2.255`.
- `-sn`: This defines our Method. It stands for "No Port Scan". It tells Nmap: "Just tell me if they are online, don't waste time scanning their ports yet."

```
(kali㉿kali)-[~]
$ nmap -sn 10.0.2.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 19:31 EST
Nmap scan report for 10.0.2.1
Host is up (0.00065s latency).
MAC Address: 52:55:0A:00:02:01 (Unknown)
Nmap scan report for 10.0.2.2
Host is up (0.00050s latency).
MAC Address: 08:00:27:88:EC:CB (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.2.3
Host is up (0.00086s latency).
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 10.0.2.15
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 2.18 seconds

(kali㉿kali)-[~]
$
```

1.2 -sn

As you can see from our previous scan, we identified the target address: `10.0.2.3`

-PE

Normally, the command `nmap -sn -PE 10.0.2.3` would be enough. However, we have a small problem.

Since our attacking machine (Kali) and the victim (Metasploitable) are on the same Local Network (we are neighbors), Nmap gets lazy. It thinks: *"Why should I bother building an ICMP packet? I'll just shout in the hallway (ARP Ping) and see if he answers."*

To learn the real technique, we need to force our Detective to do his job properly.

1. `--disable-arp-ping`: We tell Nmap: ""
2. `--packet-trace`: We tell Nmap: ""

`nmap -sn -PE 10.0.2.3 --disable-arp-ping --packet-trace`

```

(root@kali)-[/home/kali]
# nmap -sn -PE 10.0.2.3 --disable-arp-ping --packet-trace
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 21:52 EST
SENT (0.0242s) ICMP [10.0.2.15 > 10.0.2.3 Echo request (type=8/code=0) id=49513 seq=0] IP
[ttl=59 id=58397 iplen=28 ]
RCVD (0.0248s) ICMP [10.0.2.3 > 10.0.2.15 Echo reply (type=0/code=0) id=49513 seq=0] IP [t
tl=64 id=43274 iplen=28 ]
NSOCK INFO [0.0510s] nsock_iod_new2(): nsock_iod_new (IOD #1)
NSOCK INFO [0.0510s] nsock_connect_udp(): UDP connection requested to 100.100.100.100:53 (
IOD #1) EID 8
NSOCK INFO [0.0510s] nsock_read(): Read request from IOD #1 [100.100.100.100:53] (timeout:
-1ms) EID 18
NSOCK INFO [0.0510s] nsock_write(): Write request for 39 bytes to IOD #1 EID 27 [100.100.1
00.100:53]
NSOCK INFO [0.0510s] nsock_trace_handler_callback(): Callback: CONNECT SUCCESS for EID 8 [
100.100.100.100:53]
NSOCK INFO [0.0510s] nsock_trace_handler_callback(): Callback: WRITE SUCCESS for EID 27 [1
00.100.100.100:53]
NSOCK INFO [0.0890s] nsock_trace_handler_callback(): Callback: READ SUCCESS for EID 18 [10
0.100.100.100:53] (98 bytes)
NSOCK INFO [0.0890s] nsock_read(): Read request from IOD #1 [100.100.100.100:53] (timeout:
-1ms) EID 34
NSOCK INFO [0.0890s] nsock_iod_delete(): nsock_iod_delete (IOD #1)
NSOCK INFO [0.0890s] nevent_delete(): nevent_delete on event #34 (type READ)
Nmap scan report for 10.0.2.3
Host is up (0.00081s latency).
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds

(root@kali)-[/home/kali]
#

```

1.3 PE

Let's break down our first scan. The Detective rang the bell, which means we sent an **ICMP Echo request** in the **SENT** line. Then, someone actually answered back in the **RCVD** line. So we are certain that somebody lives in that building, and we can be sure of it with the final **1 host up** message.

-PP

Now, the detective tries a different approach. Instead of ringing the bell, he asks the security guard for the time to see if he reacts. We use the **-PP** flag for this. Run the command below:

```
nmap -sn -PP 10.0.2.3 --disable-arp-ping --packet-trace
```

```

(root@kali)-[/home/kali]
# nmap -sn -PP 10.0.2.3 --disable-arp-ping --packet-trace
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 21:52 EST
SENT (0.0271s) ICMP [10.0.2.15 > 10.0.2.3 Timestamp request (type=13/code=0) id=40341 seq=
0 orig=0 recv=0 trans=0] IP [ttl=41 id=2535 iplen=40 ]
RCVD (0.0281s) ICMP [10.0.2.3 > 10.0.2.15 Timestamp reply (type=14/code=0) id=40341 seq=0
orig=0 recv=10360415 trans=10360415] IP [ttl=64 id=43275 iplen=40 ]
NSOCK INFO [0.0630s] nsock_iod_new2(): nsock_iod_new (IOD #1)
NSOCK INFO [0.0640s] nsock_connect_udp(): UDP connection requested to 100.100.100.100:53 (
IOD #1) EID 8
NSOCK INFO [0.0640s] nsock_read(): Read request from IOD #1 [100.100.100.100:53] (timeout:
-1ms) EID 18
NSOCK INFO [0.0640s] nsock_write(): Write request for 39 bytes to IOD #1 EID 27 [100.100.1
00.100:53]
NSOCK INFO [0.0640s] nsock_trace_handler_callback(): Callback: CONNECT SUCCESS for EID 8 [
100.100.100.100:53]
NSOCK INFO [0.0640s] nsock_trace_handler_callback(): Callback: WRITE SUCCESS for EID 27 [1
00.100.100.100:53]
NSOCK INFO [0.1790s] nsock_trace_handler_callback(): Callback: READ SUCCESS for EID 18 [10
0.100.100.100:53] (98 bytes)
NSOCK INFO [0.1790s] nsock_read(): Read request from IOD #1 [100.100.100.100:53] (timeout:
-1ms) EID 34
NSOCK INFO [0.1790s] nsock_iod_delete(): nsock_iod_delete (IOD #1)
NSOCK INFO [0.1790s] nevent_delete(): nevent_delete on event #34 (type READ)
Nmap scan report for 10.0.2.3
Host is up (0.0017s latency).
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds

(root@kali)-[/home/kali]
#

```

1.4 PP

As we can see, we asked the security guy for the time and he answered back.

-PM

Now the detective asks the security guy for the map of the building. Let's see if this works :`nmap -sn -PM 10.0.2.3 --disable-arp-ping --packet-trace`

```

(root@kali)-[/home/kali]
# nmap -sn -PM 10.0.2.3 --disable-arp-ping --packet-trace
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 21:53 EST
SENT (0.0144s) ICMP [10.0.2.15 > 10.0.2.3 Address mask request (type=17/code=0) id=20658 s
eq=0 mask=0.0.0.0] IP [ttl=57 id=50606 iplen=32 ]
SENT (1.0173s) ICMP [10.0.2.15 > 10.0.2.3 Address mask request (type=17/code=0) id=37997 s
eq=0 mask=0.0.0.0] IP [ttl=49 id=40495 iplen=32 ]
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 2.06 seconds

(root@kali)-[/home/kali]
#

```

1.5 PM

Well, the Detective asked for the map 2 times, as we can see in the picture, but the security guy didn't answer. Even Metasploitable 2 doesn't answer this question, so we won't use it further. If the Detective used only this tool, he would have wrongly thought the building was empty.

-PS

Now we will let the detective check the building in a sneaky way. Let's throw a stone at Apartment 80. `nmap -sn -PS80 10.0.2.3 --disable-arp-ping --packet-trace`

```
(root@kali)-[/home/kali]
# nmap -sn -PS80 10.0.2.3 --disable-arp-ping --packet-trace
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 21:53 EST
SENT (0.0186s) TCP 10.0.2.15:47479 > 10.0.2.3:80 S ttl=48 id=51525 iplen=44 seq=135785568
win=1024 <mss 1460>
RCVD (0.0191s) TCP 10.0.2.3:80 > 10.0.2.15:47479 SA ttl=64 id=0 iplen=44 seq=1530532521 w
in=5840 <mss 1460>
NSOCK INFO [0.0580s] nsock_iod_new2(): nsock_iod_new (IOD #1)
NSOCK INFO [0.0580s] nsock_connect_udp(): UDP connection requested to 100.100.100.100:53 (
IOD #1) EID 8
NSOCK INFO [0.0580s] nsock_read(): Read request from IOD #1 [100.100.100.100:53] (timeout:
-1ms) EID 18
NSOCK INFO [0.0580s] nsock_write(): Write request for 39 bytes to IOD #1 EID 27 [100.100.1
00.100:53]
NSOCK INFO [0.0580s] nsock_trace_handler_callback(): Callback: CONNECT SUCCESS for EID 8 [
100.100.100.100:53]
NSOCK INFO [0.0580s] nsock_trace_handler_callback(): Callback: WRITE SUCCESS for EID 27 [1
00.100.100.100:53]
NSOCK INFO [0.0980s] nsock_trace_handler_callback(): Callback: READ SUCCESS for EID 18 [10
0.100.100.100:53] (98 bytes)
NSOCK INFO [0.0980s] nsock_read(): Read request from IOD #1 [100.100.100.100:53] (timeout:
-1ms) EID 34
NSOCK INFO [0.0980s] nsock_iod_delete(): nsock_iod_delete (IOD #1)
NSOCK INFO [0.0980s] nevent_delete(): nevent_delete on event #34 (type READ)
Nmap scan report for 10.0.2.3
Host is up (0.00085s latency).
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.12 seconds

(root@kali)-[/home/kali]
#
```

1.6 PS

As we can see, Apartment 80 answered back! So we know somebody lives in that building. Don't forget that Apt 80 can answer in two different ways:

SYN → *GREETINGS TRAVELER*, **SYN/ACK** → *WELL MET, LETS TALK!*

SYN → *GREETINGS TRAVELER*, **RST** → *LEAVE ME ALONE!*

In both situations, we are certain that somebody lives in this building.

-PA

Now, let's try the "Fake Friend" trick. We wave at the balcony and yell "Hey, thanks for the coffee yesterday!" like we know them. We use `nmap -sn -PA80 10.0.2.3 --disable-arp-ping --packet-trace`

```

(root@kali)-[/home/kali]
# nmap -sn -PA80 10.0.2.3 --disable-arp-ping --packet-trace
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 21:54 EST
SENT (0.0125s) TCP 10.0.2.15:56554 > 10.0.2.3:80 A ttl=57 id=7674 iplen=40 seq=0 win=1024

RCVD (0.0139s) TCP 10.0.2.3:80 > 10.0.2.15:56554 R ttl=64 id=0 iplen=40 seq=1988159265 wi
n=0
NSOCK INFO [0.0480s] nsock_iod_new2(): nsock_iod_new (IOD #1)
NSOCK INFO [0.0480s] nsock_connect_udp(): UDP connection requested to 100.100.100.100:53 (
IOD #1) EID 8
NSOCK INFO [0.0480s] nsock_read(): Read request from IOD #1 [100.100.100.100:53] (timeout:
-1ms) EID 18
NSOCK INFO [0.0480s] nsock_write(): Write request for 39 bytes to IOD #1 EID 27 [100.100.1
00.100:53]
NSOCK INFO [0.0480s] nsock_trace_handler_callback(): Callback: CONNECT SUCCESS for EID 8 [
100.100.100.100:53]
NSOCK INFO [0.0480s] nsock_trace_handler_callback(): Callback: WRITE SUCCESS for EID 27 [1
00.100.100.100:53]
NSOCK INFO [0.0860s] nsock_trace_handler_callback(): Callback: READ SUCCESS for EID 18 [1
0.100.100.100:53] (98 bytes)
NSOCK INFO [0.0860s] nsock_read(): Read request from IOD #1 [100.100.100.100:53] (timeout:
-1ms) EID 34
NSOCK INFO [0.0860s] nsock_iod_delete(): nsock_iod_delete (IOD #1)
NSOCK INFO [0.0860s] nevent_delete(): nevent_delete on event #34 (type READ)
Nmap scan report for 10.0.2.3
Host is up (0.0015s latency).
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds

(root@kali)-[/home/kali]
# █

```

1.7 SA

We are acting like we already have a connection (ACK). The target gets confused because we never met, so they yell back “Who are you? I didn’t send you anything!” This response is called a RST (Reset) packet.

If we get a RST back, it means someone is definitely home.

-PU

Now we are going to use `nmap -sn -PU53 10.0.2.3 --disable-arp-ping --packet-trace`

```

(root@kali)-[/home/kali]
# nmap -sn -PU53 10.0.2.3 --disable-arp-ping --packet-trace
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 21:54 EST
SENT (0.0920s) UDP 10.0.2.15:55740 > 10.0.2.3:53 ttl=46 id=55899 iplen=58
SENT (0.0922s) UDP 10.0.2.15:55740 > 10.0.2.3:53 ttl=55 id=55899 iplen=40
SENT (0.0923s) UDP 10.0.2.15:55740 > 10.0.2.3:53 ttl=45 id=55899 iplen=74
RCVD (0.0937s) UDP 10.0.2.3:53 > 10.0.2.15:55740 ttl=64 id=0 iplen=90
NSOCK INFO [0.1280s] nssock_iod_new2(): nssock_iod_new (IOD #1)
NSOCK INFO [0.1280s] nssock_connect_udp(): UDP connection requested to 100.100.100.100:53 (
IOD #1) EID 8
NSOCK INFO [0.1280s] nssock_read(): Read request from IOD #1 [100.100.100.100:53] (timeout:
-1ms) EID 18
NSOCK INFO [0.1280s] nssock_write(): Write request for 39 bytes to IOD #1 EID 27 [100.100.1
00.100:53]
NSOCK INFO [0.1280s] nssock_trace_handler_callback(): Callback: CONNECT SUCCESS for EID 8 [
100.100.100.100:53]
NSOCK INFO [0.1280s] nssock_trace_handler_callback(): Callback: WRITE SUCCESS for EID 27 [1
00.100.100.100:53]
NSOCK INFO [0.1690s] nssock_trace_handler_callback(): Callback: READ SUCCESS for EID 18 [10
0.100.100.100:53] (98 bytes)
NSOCK INFO [0.1690s] nssock_read(): Read request from IOD #1 [100.100.100.100:53] (timeout:
-1ms) EID 34
NSOCK INFO [0.1690s] nssock_iod_delete(): nssock_iod_delete (IOD #1)
NSOCK INFO [0.1690s] nevent_delete(): nevent_delete on event #34 (type READ)
Nmap scan report for 10.0.2.3
Host is up (0.0017s latency).
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds

(root@kali)-[/home/kali]
#

```

1.8 PU

We assumed the garage (UDP Port) was locked and we were waiting for the owner to come out and yell at us (“ICMP Port Unreachable”).

But look at the screenshot closely.

- A UDP packet to Port 53 (DNS).
- A UDP packet back! (Not an error message).

Wait, doesn’t UDP have no handshake? How did it reply?

Well I was surprised as much as you. So I investigated further. UDP doesn’t shake hands (SYN/ACK). But turns out Nmap is smarter than we think. When probing Port 53, Nmap doesn’t just send an empty box. It sends a specific question (like “Who is Google?”). Since Port 53 is actually OPEN on our target (Metasploitable), the service inside caught the question and immediately answered back. After all we found the host is up again!

-PR

Now let’s check our last tool. Since we are on the same local network, we can simply remove the `--disable-arp-ping` section from our code and let Nmap do its natural thing.

```
nmap -sn -PR 10.0.2.3 --packet-trace
```

```

(root@kali)-[/home/kali]
# nmap -sn -PR 10.0.2.3 --packet-trace
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 21:55 EST
SENT (0.0160s) ARP who-has 10.0.2.3 tell 10.0.2.15
RCVD (0.0170s) ARP reply 10.0.2.3 is-at 08:00:27:74:F4:39
NSOCK INFO [0.0500s] nssock_iod_new2(): nssock_iod_new (IOD #1)
NSOCK INFO [0.0510s] nssock_connect_udp(): UDP connection requested to 100.100.100.100:53 (
IOD #1) EID 8
NSOCK INFO [0.0510s] nssock_read(): Read request from IOD #1 [100.100.100.100:53] (timeout:
-1ms) EID 18
NSOCK INFO [0.0510s] nssock_write(): Write request for 39 bytes to IOD #1 EID 27 [100.100.1
00.100:53]
NSOCK INFO [0.0510s] nssock_trace_handler_callback(): Callback: CONNECT SUCCESS for EID 8 [
100.100.100.100:53]
NSOCK INFO [0.0510s] nssock_trace_handler_callback(): Callback: WRITE SUCCESS for EID 27 [1
00.100.100.100:53]
NSOCK INFO [0.0930s] nssock_trace_handler_callback(): Callback: READ SUCCESS for EID 18 [10
0.100.100.100:53] (98 bytes)
NSOCK INFO [0.0930s] nssock_read(): Read request from IOD #1 [100.100.100.100:53] (timeout:
-1ms) EID 34
NSOCK INFO [0.0930s] nssock_iod_delete(): nssock_iod_delete (IOD #1)
NSOCK INFO [0.0930s] nevent_delete(): nevent_delete on event #34 (type READ)
Nmap scan report for 10.0.2.3
Host is up (0.00098s latency).
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.11 seconds

(root@kali)-[/home/kali]
# █

```

Again, the host is up. So, everything worked perfectly except for the **-PM**. Our Detective is finally satisfied with the results. We know the target is there and now he is ready to investigate further.

Story Time, Ch2

The Detective lowered his binoculars. The rain was still pouring down on Kali Linux City, but a smirk appeared on his face.

The initial investigation was a success. The tricks worked.

- The doorbell was rung (**-PE**).
- The time was asked (**-PP**)
- The stone was thrown (**-PS**).
- The fake wave was made (**-PA**).
- Even the garage door was checked (**-PU**).

All evidence pointed to one single location: 10.0.2.3.

He stood directly in front of the building now. The lights were on. The target was definitely inside. But knowing where the Smiling Bandit lives is not enough to catch him. He had to get inside.

The Detective looked up at the massive apartment complex. It wasn't just a simple house; it was a fortress with 65,535 doors(Ports). Most of them were locked tight. Some might be booby-trapped with firewalls. But somewhere, on some floor, the Bandit left a window cracked open.

"Phase 1: Discovery is complete," the Detective whispered to his radio. "Initiating Phase 2: The Port Scan."

He walked up to the first door. It was time to start knocking.

Phase 2: The Port Scan

Now, we need to check the doors (ports) to investigate further. If we just let the Detective do his thing without giving him any specific orders (no parameters), he will follow his standard routine: checking the most popular 1000 ports.

```
(root@kali)-[/home/kali]
# nmap 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 21:55 EST
Nmap scan report for 10.0.2.3
Host is up (0.00080s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.56 seconds

(root@kali)-[/home/kali]
#
```

2.0 nmap

`nmap 10.0.2.3`

As we can see from the picture, the Detective checked the top 1000 ports and listed every open one. It worked, but it's not enough. We want to control exactly how he knocks and which doors he checks. We want to lead the way with parameters.

So, let's dive in.

1. -sS (TCP SYN Scan) aka "The Ghost": When we use this, we send a **SYN** to a port. If we receive a **SYN/ACK**, we immediately send a **RST**.

- The Detective knocks on the door, and the moment someone opens it, he runs away. Since he never actually "enters" (doesn't complete the connection), he leaves almost no trace in the logs.
- `SYN → SYN/ACK → RST`

2. -sT (TCP Connect Scan) aka "The Polite Guy": The Detective knocks on the door again. But this time, he won't run away. When the door opens, he enters, says "Hello, I am the Detective," shakes hands, and then ends the conversation properly.

- It works, but it's noisy. Since you completed the full connection, the target logs your visit. We usually only use this if we don't have root privileges.
- `SYN → SYN/ACK → ACK → FIN`

3. -sU (UDP Scan) aka "The Shouting Man": Here, we approach a window or a dark room and yell inside.

- If someone yells "GET OUT!" (`ICMP Unreachable`), the port is Closed.
- If they answer back (like in our earlier `-PU` scan), the port is Open.
- If there is total silence... well, it's either Open or Filtered (blocked by a firewall). The person inside might be there, but they are just ignoring us.

Okay, we know the theory. Now let's help the Detective apply these new skills in the field. First, we tell the Detective to use his "Ghost Mode" (`-sS`) to stay stealthy. Also, we are in a hurry, so we tell him to only check the top 100 most popular ports (`-F`) instead of the usual 1000.

`sudo nmap -sS -F 10.0.2.3`

```

(root@kali)-[/home/kali]
# nmap -sS -F 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 22:14 EST
Nmap scan report for 10.0.2.3
Host is up (0.0016s latency).
Not shown: 82 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
513/tcp   open  login
514/tcp   open  shell
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
8009/tcp  open  ajp13
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.18 seconds

(root@kali)-[/home/kali]
# █

```

2.1 sS

Now, let's tell our Detective to be the "Polite Guy." He needs to knock, shake hands and establish a full connection (-sT). But this time, we don't want to check everywhere. We suspect the Bandit is hiding in a specific corridor, so we tell him: *"Only check the doors from 3000 to 4000."*

`nmap -sT -p3000-4000 10.0.2.3`

```

(root@kali)-[/home/kali]
# nmap -sT -p3000-4000 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 22:17 EST
Nmap scan report for 10.0.2.3
Host is up (0.0027s latency).
Not shown: 999 closed tcp ports (conn-refused)
PORT      STATE SERVICE
3306/tcp  open  mysql
3632/tcp  open  distccd
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.53 seconds

(root@kali)-[/home/kali]
# █

```

2.2 sT

The Detective found a new open door: Port 3632. He notes it down.

Now, let's check the services that don't use doors (UDP). We tell him to shout into the dark rooms to see if anyone answers.

Warning: Since he has to wait for a response (or lack of one), this scan takes a long time. To keep it fast, we will stick to the top 100 ports (-F) again. `nmap -F -sU 10.0.2.3`

```
(root@kali)-[/home/kali]
# nmap -sU -F 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 22:19 EST
Stats: 0:00:01 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 14.00% done; ETC: 22:19 (0:00:12 remaining)
Stats: 0:00:26 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 40.78% done; ETC: 22:20 (0:00:39 remaining)
Stats: 0:00:42 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 56.22% done; ETC: 22:20 (0:00:33 remaining)
Stats: 0:00:56 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 67.89% done; ETC: 22:20 (0:00:27 remaining)
Nmap scan report for 10.0.2.3
Host is up (0.0011s latency).
Not shown: 93 closed udp ports (port-unreach)
PORT      STATE      SERVICE
53/udp    open       domain
68/udp    open|filtered dhcpc
69/udp    open|filtered tftp
111/udp   open       rpcbind
137/udp   open       netbios-ns
138/udp   open|filtered netbios-dgm
2049/udp  open       nfs
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 97.37 seconds

(root@kali)-[/home/kali]
#
```

2.3 sU

The Detective is done playing games. He knows the target is inside, and he's tired of guessing. He decides to search the entire building, from the basement to the penthouse. Every. Single. Room.

But he's not just going to walk; he's going to sprint.

Here is the monster command we used:

```
nmap -sS -p- -r -T3 --min-rate 1000 --max-rate 5000 --min-parallelism 100
10.0.2.3 -v
```

Let's decode the Detective's new strategy:

- **-p-** Normally, Nmap only checks the top 1000 popular doors. But the Smiling Bandit might be hiding in a secret room like Room 31337 or Room 50146. So we tell the Detective: "Check all 65,535 doors. Don't skip a single one."
- **-r** The Detective jumps around randomly (Door 80, then Door 5000, then Door 22) to confuse the security guards. With **-r**: We tell him: "Be systematic. Check Door 1, then Door 2, then Door 3..." We want a perfect, orderly list.

- We injected the Detective with adrenaline. `--min-rate 1000`: "You must knock on at least 1000 doors per second!" `--min-parallelism 100`: "Use your shadow clones! Check 100 doors at the exact same time." `--max-rate 5000`: But don't knock more than 5000 doors per second!
- `-v` The Detective uses his walkie-talkie. Instead of waiting until the end to give a report, he shouts immediately every time he finds an open door:

```

(root@kali)-[/home/kali]
# nmap -sS -p- -r -T3 --min-rate 1000 --max-rate 5000 --min-parallelism 100 10.0.2.3 -v
v
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 22:26 EST
Initiating ARP Ping Scan at 22:26
Scanning 10.0.2.3 [1 port]
Completed ARP Ping Scan at 22:26, 0.05s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 22:26
Completed Parallel DNS resolution of 1 host. at 22:26, 0.04s elapsed
Initiating SYN Stealth Scan at 22:26
Scanning 10.0.2.3 [65535 ports]
Discovered open port 21/tcp on 10.0.2.3
Discovered open port 22/tcp on 10.0.2.3
Discovered open port 23/tcp on 10.0.2.3
Discovered open port 25/tcp on 10.0.2.3
Discovered open port 53/tcp on 10.0.2.3
Discovered open port 80/tcp on 10.0.2.3
Discovered open port 111/tcp on 10.0.2.3
Discovered open port 139/tcp on 10.0.2.3
Discovered open port 445/tcp on 10.0.2.3
Discovered open port 512/tcp on 10.0.2.3
Discovered open port 513/tcp on 10.0.2.3
Discovered open port 514/tcp on 10.0.2.3
Discovered open port 1099/tcp on 10.0.2.3
Discovered open port 1524/tcp on 10.0.2.3
Discovered open port 2049/tcp on 10.0.2.3
Discovered open port 2121/tcp on 10.0.2.3
Discovered open port 3306/tcp on 10.0.2.3
Discovered open port 3632/tcp on 10.0.2.3
Discovered open port 5432/tcp on 10.0.2.3
Discovered open port 5900/tcp on 10.0.2.3
Discovered open port 6000/tcp on 10.0.2.3
Discovered open port 6667/tcp on 10.0.2.3
Discovered open port 6697/tcp on 10.0.2.3
Discovered open port 8009/tcp on 10.0.2.3
Discovered open port 8180/tcp on 10.0.2.3
Discovered open port 8787/tcp on 10.0.2.3
Discovered open port 39500/tcp on 10.0.2.3
Discovered open port 41314/tcp on 10.0.2.3
Discovered open port 45826/tcp on 10.0.2.3
Discovered open port 50146/tcp on 10.0.2.3
Completed SYN Stealth Scan at 22:26, 26.59s elapsed (65535 total ports)
Nmap scan report for 10.0.2.3
Host is up, received arp-response (0.00044s latency).
Scanned at 2026-01-06 22:26:23 EST for 26s
Not shown: 65505 closed tcp ports (reset)

```

PORT	STATE	SERVICE	REASON
21/tcp	open	ftp	syn-ack ttl 64
22/tcp	open	ssh	syn-ack ttl 64
23/tcp	open	telnet	syn-ack ttl 64
25/tcp	open	smtp	syn-ack ttl 64
53/tcp	open	domain	syn-ack ttl 64
80/tcp	open	http	syn-ack ttl 64
111/tcp	open	rpcbind	syn-ack ttl 64
139/tcp	open	netbios-ssn	syn-ack ttl 64
445/tcp	open	microsoft-ds	syn-ack ttl 64
512/tcp	open	exec	syn-ack ttl 64
513/tcp	open	login	syn-ack ttl 64
514/tcp	open	shell	syn-ack ttl 64
1099/tcp	open	rmiregistry	syn-ack ttl 64
1524/tcp	open	ingreslock	syn-ack ttl 64
2049/tcp	open	nfs	syn-ack ttl 64
2121/tcp	open	ccproxy-ftp	syn-ack ttl 64
3306/tcp	open	mysql	syn-ack ttl 64
3632/tcp	open	distccd	syn-ack ttl 64
5432/tcp	open	postgresql	syn-ack ttl 64
5900/tcp	open	vnc	syn-ack ttl 64
6000/tcp	open	X11	syn-ack ttl 64
6667/tcp	open	irc	syn-ack ttl 64
6697/tcp	open	ircs-u	syn-ack ttl 64
8009/tcp	open	ajp13	syn-ack ttl 64
8180/tcp	open	unknown	syn-ack ttl 64
8787/tcp	open	msgsrvr	syn-ack ttl 64
39500/tcp	open	unknown	syn-ack ttl 64

```
41314/tcp open  unknown      syn-ack ttl 64
45826/tcp open  unknown      syn-ack ttl 64
50146/tcp open  unknown      syn-ack ttl 64
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Read data files from: /usr/share/nmap
Nmap done: 1 IP address (1 host up) scanned in 26.77 seconds
Raw packets sent: 65536 (2.884MB) | Rcvd: 65536 (2.622MB)

(root@kali)-[/home/kali]
#
```

2.4 monster

Look at the evidence list! We found dozens of open doors that a normal scan might have missed. The Smiling Bandit has nowhere left to hide.

Story Time, Ch3

The Detective wiped the rain from his face and took a deep breath. He was exhausted. He had just checked 65,535 doors in less than half a minute. He moved like a blur, a phantom sprinting through the hallways.

He leaned against a streetlamp and opened his notebook. He expected to find a fortress. He expected heavy steel doors, barred windows, and armed guards.

But what he saw made him shake his head.

This wasn't a fortress. It was a public park.

- The delivery door was wide open.
- The main glass doors were unlocked, welcoming anyone who walked by.
- The door to the files was slightly ajar.
- Even the high-level windows were left unlatched.

"The Bandit is careless," the Detective whispered, looking up at the lit windows. "He didn't just leave a window open; he took the whole wall down."

We know the doors are open. But now the real question is: Who is standing behind them? Is it a tough security guard? A sleeping old man? Or a confused rookie?

The Detective straightened his tie. It was time to stop knocking and start talking.

"Phase 3," he noted in his journal. "Interrogation (Version Detection)."

PHASE 3 Interrigation (Version Detection)

The Detective is now standing inside the lobby. He sees dozens of open doors. But an open door means nothing if you don't know who is waiting behind it. Is Port 21 guarded by a tough, updated security guard? Or is it guarded by an old, sleepy, vulnerable old man? To find out, the Detective can't just knock. He needs to talk. He needs to check their IDs. This is called Service Version Detection with `-sV`.

```
(root@kali)-[/home/kali]
# sudo nmap -sV -T3 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 22:57 EST
Nmap scan report for 10.0.2.3
Host is up (0.00055s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.91 seconds

(root@kali)-[/home/kali]
#
```

3.1 sV

when we write `nmap -sV -T3 10.0.2.3` we can see that there is another column in the CLI. This is called version scan. Now we now what version use every every ports. And we found it. The criminal Mr. vsftpd 2.3.4 is in the apartment number 21.

Let's check it with `--version-all` parameter. We can use `--version-all` or `--version-light` to let detective asks more questions.

```
(root@kali)-[/home/kali]
# sudo nmap -sV -T3 -p21 10.0.2.3 --version-all
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 23:19 EST
Nmap scan report for 10.0.2.3
Host is up (0.0017s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.44 seconds

(root@kali)-[/home/kali]
#
```

3.2 sV2

Still the same answer. But nonetheless we found out the suspicious person. Lets ask more questions.

```
(root@kali)-[/home/kali]
# sudo nmap -sV -T3 -p21 -O 10.0.2.3 --version-all
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 23:22 EST
Nmap scan report for 10.0.2.3
Host is up (0.0014s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: OS: Unix

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.67 seconds

(root@kali)-[/home/kali]
#
```

3.3

`nmap -sV -T3 -p21 -O 10.0.2.3 --version-all`

- **-O** The Detective checks the building's foundation to see if it's built on Windows or Linux.
- **--traceroute** He draws a map of the road, marking every intersection (router) between us and the target.
- **-sC** He uses his standard toolkit (flashlight & magnifying glass) to spot extra details, like unlocked handles or hidden notes.
- **-A** Instead of typing **-sV -sC -O --traceroute** separately, use **-A**. It tells the Detective to do ALL of the above at once. It's loud, but it gets everything you need in one go.

The Detective has finished the job. He knows where the Bandit is. But as the old saying goes: *"Spoken words fly away, written words remain."*

You, the Chief, need a report on your desk. But how should he write it?

- **-oN** The Detective types up a classic report on his typewriter. It is designed for human eyes. It's clean, structured, and easy for you to read while sipping coffee. (Standard text file).
- **-oG** The Detective condenses the entire investigation into one single line. This isn't for you; it's for the robots. If you want to feed this data into other scripts or search through it quickly using tools like **grep**, this is the format.
- **-oX** The Detective formats the data perfectly for the digital archives. Think of this like an Excel sheet or a database entry. It allows other hacking tools (like Metasploit) to easily import and understand the findings.
- **-oA** The Detective takes no chances. He prints three copies of the report (Normal, Grepable, and XML) and drops the whole stack on your desk. This is the pro move. You never know which format you'll need later, so we always ask for all of them.

```

(root@kali)-[/home/kali]
# nmap -A -p- -T4 -oN scan_results.txt 10.0.2.3
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-06 23:33 EST
Stats: 0:02:02 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 96.67% done; ETC: 23:35 (0:00:03 remaining)
Stats: 0:02:19 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 96.67% done; ETC: 23:35 (0:00:04 remaining)
Nmap scan report for 10.0.2.3
Host is up (0.0010s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 10.0.2.15
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-date: 2026-01-07T04:35:54+00:00; -4s from scanner time.
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
|_sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
53/tcp    open  domain       ISC BIND 9.4.2
|_dns-nsid:
|_bind.version: 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp   open  rpcbind      2 (RPC #100000)
|_rpcinfo:
|   program version    port/proto  service
|   100000  2                111/tcp    rpcbind
|   100000  2                111/udp    rpcbind
|   100003  2,3,4           2049/tcp    nfs
|   100003  2,3,4           2049/udp    nfs
|   100005  1,2,3           36691/udp   mountd
|   100005  1,2,3           45826/tcp   mountd
|   100021  1,3,4           41314/tcp   nlockmgr
|   100021  1,3,4           57129/udp   nlockmgr
|   100024  1                39500/tcp   status
|   100024  1                59363/udp   status
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
|_mysql-info:

```

```

| Protocol: 10
| Version: 5.0.51a-3ubuntu5
| Thread ID: 13
| Capabilities flags: 43564
| Some Capabilities: Support41Auth, SupportsTransactions, SwitchToSSLAfterHandshake, SupportsComp
| resson, ConnectWithDatabase, Speaks41ProtocolNew, LongColumnFlag
| Status: Autocommit
|_ Salt: 8y~+XZ:RoZQg5pV<i~]8
3632/tcp open distccd distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceNa
| me=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
|_ ssl-date: 2026-01-07T04:35:54+00:00; -5s from scanner time.
5900/tcp open vnc VNC (protocol 3.3)
| vnc-info:
| Protocol version: 3.3
| Security types:
|_ VNC Authentication (2)
6000/tcp open X11 (access denied)
6667/tcp open irc UnrealIRCd
| irc-info:
| users: 2
| servers: 1
| lusers: 2
| lservers: 0
| server: irc.Metasploitable.LAN
| version: Unreal3.2.8.1. irc.Metasploitable.LAN
| uptime: 0 days, 6:56:30
| source ident: nmap
| source host: C29CBC04.EB72D3BE.7B559A54.IP
|_ error: Closing Link: xkbwrczth[10.0.2.15] (Quit: xkbwrczth)
6697/tcp open irc UnrealIRCd
8009/tcp open ajp13 Apache Jserv (Protocol v1.3)
|_ ajp-methods: Failed to get a valid response for the OPTION request

```

```

|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http      Apache Tomcat/Coyote JSP engine 1.1
|_http-favicon: Apache Tomcat
|_http-title: Apache Tomcat/5.5
|_http-server-header: Apache-Coyote/1.1
8787/tcp open  drb       Ruby DRb RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbc)
39500/tcp open status    1 (RPC #100024)
41314/tcp open nlockmgr  1-4 (RPC #100021)
45826/tcp open mountd    1-3 (RPC #100005)
50146/tcp open java-rmi  GNU Classpath grmiregistry
MAC Address: 08:00:27:74:F4:39 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_smb2-time: Protocol negotiation failed (SMB2)
|_smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_ System time: 2026-01-06T23:35:45-05:00
|_clock-skew: mean: 1h14m55s, deviation: 2h29m59s, median: -5s
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)

TRACEROUTE
HOP RTT      ADDRESS
1   1.03 ms  10.0.2.3

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/
.
Nmap done: 1 IP address (1 host up) scanned in 160.54 seconds

(root@kali)-[/home/kali]
#

```

3.5

So, putting everything we learned together, here is the “Gold Standard” command for the Detective. We tell him to check every single room (**-p-**), use all his aggressive tools (**-A**), move fast (**-T4**), and save the evidence for the Chief (**-oN**).

```
nmap -A -p- -T4 -oN scan_result.txt 10.0.2.3
```

Now, we have a full dossier on the Smiling Bandit. The investigation phase is complete. It’s time to exploit.

The Detective’s Secret Toolkit (Evasion & Stealth)

Sometimes, the building is guarded by a tough bouncer known as The Firewall. He blocks normal knocks (**-ss**, **-st**) and refuses to let anyone in.

To get past him, the Detective has to get creative. He stops acting like a policeman and starts acting like a magician.

Note: For this specific investigation (Metasploitable), the target is careless and the doors are wide open, so we don't need to use these tricks right now. However, every Master Detective must keep these moves in their back pocket for the day they face a real fortress.

Here are the tricks we keep in reserve:

1- Null Scan (-sN): The Detective stands in front of the door and says absolutely nothing (No flags).

- The guard gets annoyed and yells "Get lost!" (RST).
- The guard stands there confused and silent (No response).
- If you write `nmap -sN 10.0.2.3`, you'll realize everything is open|filtered since there is no firewall in metasploitable2.

2- FIN Scan (-sF): The Detective walks up and immediately says "Goodbye!" (FIN flag) without ever saying hello.

It's a trick to see if the system reacts to an impossible conversation.

3- Xmas Scan (-sX): The Detective lights up like a Christmas Tree. He turns on all the blinking lights (FIN, URG, PSH flags).

- It's meant to overwhelm the guard and see if he slips up.
- The name itself is really Xmas Scan because in Wireshark it looks like a Xmas Tree.

3- ACK Scan (-sA): This isn't for finding open doors. It's for finding the firewall.

- The Detective acts like he is already inside. He sends an "Okay, got it" (ACK) packet.
- He wants to see if a Firewall blocks him or if the message reaches the door. It helps him map out the security perimeters.

4- Decoy Scan (-D)

- The Detective doesn't go alone. He brings 10 other people wearing the exact same trench coat (`nmap -D RND:10 ...`).
- The security guard sees 11 people knocking at the same time. He doesn't know which one is the real Detective and which ones are fake. The logs get flooded with fake IPs.

5- Fragmentation (-f)

- The Detective writes his ID on a piece of paper, tears it into tiny confetti pieces (8 bytes), and throws them through the mail slot one by one.
- The security scanner at the door can't read the tiny pieces. But once they get inside, the target tapes them back together and reads the message. A classic way to bypass inspection.

6- Idle/Zombie Scan (-sI)

- The Detective sits in a café across the street. He hacks an innocent bystander's phone (The Zombie) and uses their phone to call the target.
- If the police trace the call, it leads to the innocent Zombie, not the Detective. This is the ultimate stealth.

7- IP Spoofing (-S)

Here, the Detective doesn't just hide; he commits identity theft. He puts on a mask that looks exactly like the CEO or a Trusted Neighbor.

- The Detective knocks on the door, but his badge says he is "192.168.1.55" (someone else), not his real address. If the building's security has a rule like this trick bypasses it. Here is the problem. When the person inside answers, they send the invitation to the REAL 192.168.1.55, not to the detective. So, we usually don't hear the answer. We are knocking blindly just to confuse the logs or trigger a reaction elsewhere.
- In Decoy, we are in the crowd, so we get the answer. In Spoofing (-S), we are pretending to be someone else entirely, so the answer goes to them.

8- No Ping(-Pn)

Normally, before the Detective approaches a house, he looks to see if the lights are on (Ping). If the house looks dark and silent, he thinks, "Nobody is home," and walks away to save time.

But some criminals are smart. They sit inside the dark, holding their breath, pretending to be away (Blocking Ping/ICMP requests).

-Pn tells the Detective: *"I don't care if the lights are off. I don't care if it looks abandoned. I know he is in there. Skip the check and start knocking on the doors immediately."*

If you are 100% sure the IP address is correct but Nmap refuses to scan, use **-Pn**.

Story Time, Ch 4

The Detective sat back in his car, the rain drumming against the roof. He looked at the passenger seat where his Secret Toolkit lay untouched.

He hadn't needed the masks (-S), the crowd of decoys (-D), or the zombie phones (-SI) tonight. The building was weak. The Smiling Bandit had gotten arrogant, leaving the front door (Port 21) unlocked and unguarded.

But the Detective knew that not every case would be this easy. One day, he would face a bank, a government server, a paranoid sysadmin. And on that day, those secret tools would be the only way in.

For now, though, the reconnaissance was over.

He picked up his radio and dialed the Chief. “Chief, I have a visual. The target is confirmed at 10.0.2.3.”

“Good work, Detective,” the Chief’s voice crackled.

“Is the perimeter secure?”

“Negative. It’s full of holes. I found the Smiling Bandit hiding behind Port 21. He is running vsftpd 2.3.4, and he’s vulnerable.”

There was a pause on the other end.

“Excellent. Pack up your gear, Detective. You’ve done your part. We are sending in the Assault Team.”

The Detective smirked and started the engine. The scanning phase was done. It was time to stop knocking. It was time to kick the door down.

Next Episode: The Raid (Exploiting with Metasploit)

We have completed the Reconnaissance Phase. The Detective has done his job perfectly. He identified the target. He found the open door (Port 21). He even identified the criminal hiding inside (vsftpd 2.3.4).

But finding a vulnerability is useless if you don’t know how to use it.

In the next episode of we are trading our magnifying glass for a Battering Ram.

- We will open the Metasploit Framework.
- We will load the weapon (Exploit) designed specifically for the “Smiling Bandit.”
- We will trigger the Backdoor and take full control of the system (Root Access).

The Detective is going home. The Assault Team is moving in.

You don’t want to miss the breach. Follow me to get notified when the Raid begins!

Author’s Note:

The technical concepts and the “Detective” narrative structure are my own original work. I utilized AI assistance to refine the language and enhance the metaphors/analogies, ensuring the storytelling is as engaging and clear as possible.